

Take-Home Case: AI Support Draft Engine

Role: Operational Engineer **Stage:** Second-round interview **Format:** Prepared at home, presented live **Expected effort:** ~4 hours max (please don't over-invest — we're evaluating judgment, not polish)

A note on how this fits: This presentation *is* your second-round interview — not extra homework you do to earn one. We've kept the time box deliberately short and told you exactly what we're scoring on. You're designing against illustrative sample data, not our production systems, and there's no expectation of a finished or production-ready build.

Context

We're a regulated fintech. Every day our customer success team receives customer messages, primarily by email. We want to build a system that **automatically generates a drafted reply whenever a customer writes to us** — a draft that a human reviews before anything is sent.

We'd like you to design that system end-to-end and present your thinking. The goal of this case is to see **how you reason about building with AI in a production, regulated environment: not just whether it works, but whether it's safe, measurable, and operable.**

Who you'll be working with matters here. This system is built *for* and *with* our customer success people — not engineers. A big part of the job is taking something genuinely complex and explaining it clearly and simply to non-technical colleagues who need to trust it, use it, and give feedback on it. We'll be paying close attention to how clearly you can articulate your design to a mixed audience. **Don't assume the room is technical.**

A realistic set of sample customer inquiries is included at the end of this brief. Please use them as the concrete basis for your design and demo.

The challenge

Design a system that turns an incoming customer message into a high-quality drafted reply for a support agent to review.

Assume:

- ~20 messages/day today — but we're growing fast, so design something that comfortably handles **200/day**
- Channel: **email only** (we're leaving live chat out of scope for now)
- **Language: most messages come in Danish, some in English** — your system has to handle Danish natively and reply in the customer's language
- Emails arrive as **tickets in HubSpot**, which is our CRM and support tool — your system should fit into the HubSpot ticket workflow (e.g. reading the incoming ticket, attaching or surfacing the draft for the agent there)
- A regulated fintech context (think account balances, fees, KYC, disputes)
- An existing help center and a history of resolved tickets you can draw on
- A customer success team that today writes every reply by hand

The brief is deliberately open. Part of what we're looking for is **how you scope an ambiguous problem before you start designing.**

What to deliver

A **presentation** (slides) explaining the architecture and your approach.

Bonus points for a working prototype. If you want to go further, bring a small repo or notebook that actually drafts replies to the sample inquiries below. This is genuinely optional and sits *above and beyond* the 4-hour expectation — we know a real prototype takes longer. A strong design with no prototype scores well; a prototype on top is extra credit and a great way to stand out. It does not need to be production-ready, and not building one won't count against you.

Please structure your presentation around these sections so we can compare candidates fairly:

1. **Scope & assumptions** — how you framed the problem and what you chose to leave out
2. **Architecture** — the end-to-end pipeline from incoming message to reviewed draft
3. **Grounding & guardrails** — how drafts stay accurate and safe in a regulated context
4. **Rollout plan** — how you'd ship this without breaking trust
5. **Measurement** — how you'd know it's working, before and after launch
6. **Self-reinforcing loop** — how the system gets *better over time* on its own (see below)
7. **Demo (bonus)** — a working prototype drafting replies to the sample inquiries, if you built one
8. **What's next** — what you'd do with another week, and the biggest risks you see

On the self-reinforcing loop

Every time an agent reviews a draft, they generate a signal: they accept it, lightly edit it, rewrite it, or discard it. We want to see how you'd turn that everyday signal into a system that **improves itself over time** rather than staying static.

We're not looking for one "right" answer — we want your reasoning. Things you might address: what signal you capture from each agent action (the final sent reply, the edit, an explicit thumbs-up/down); where it's stored; and how it flows back to improve drafts — for example by feeding good resolved tickets back into retrieval, building a growing library of approved answers, updating prompts or few-shot examples, flagging recurring questions that should become help-center articles, or (eventually) fine-tuning. Just as important: how you'd avoid the loop reinforcing *bad* habits, and how a human stays in control of what the system learns.

What we're looking for

We'll score each candidate on the following. The strongest signals are usually **measurement** and **knowing when *not* to draft** — these separate engineers who design the happy path from those who design for the day the model is confidently wrong.

Scoping & judgment. Did you ask the right questions and make sensible assumptions instead of designing blindly? Did you cut scope deliberately?

Architecture. A clear pipeline: ingest → triage/classify → retrieve context → generate → human review → send → capture the agent's edit. We're looking for sound, simple choices over impressive-but-fragile ones.

Grounding. Are drafts anchored in retrieved sources (help docs, past resolved tickets, account data) rather than free-generated? Can the system show *why* it said what it said?

Guardrails & failure modes. How do you prevent hallucinated balances, wrong fees, or accidental financial/legal advice? We want to see PII handling, escalation paths for compliance-sensitive topics, output validation, and a clear answer to: *"the model is 80% confident about a fee question — what happens?"*

Human-in-the-loop. When does the system draft, when does it stay silent and route to a human, and when (if ever) does it auto-send? In our context, drafts should not auto-send on day one.

Rollout. A phased plan: shadow mode (draft silently, compare to what agents actually sent) → suggest mode (agents see and edit drafts) → measure → only then consider auto-send for narrow, safe categories.

Measurement. A real plan to evaluate quality: an offline eval set of tickets with known-good answers, agent acceptance/edit rate as the core online metric, CSAT, and LLM-as-judge with human spot-checks. If you can't measure it, you can't operate it.

Self-reinforcing loop. Does the design get better over time from agent feedback instead of staying static? We want to see a concrete path from "agent edited this draft" to "the next draft is better" — and an awareness of how to stop the loop from learning bad habits.

HubSpot fit. Does the design realistically slot into the existing HubSpot ticket workflow rather than assuming a greenfield system?

Operational thinking. Monitoring for quality drift, cost per ticket and how to control it, latency, and what happens when the model provider has an outage. This is the operational role — we care about this a lot.

Communication (weighted heavily). Can you take something complex and make it clear and simple for a non-technical audience? You'll be working day-to-day with customer success colleagues, so this matters as much as the design itself. Can you explain *why* a choice was made, and what the trade-off was, without hiding behind jargon? A candidate who designs a modest system and explains it brilliantly will score higher than one who designs an elaborate system no one in the room can follow.

Working prototype (bonus). Not required, and its absence won't count against you — but if you bring one that actually drafts replies to the sample inquiries, it's a strong positive signal that you can move from design to something real.

We are **not** scoring you on model novelty, UI polish, or how many features you cram in. A focused, safe, measurable design that you can explain clearly beats an ambitious one every time. We don't expect deep production experience — we care most about clear thinking, sound judgment, and clear communication.

Logistics

- Send your slides (and link, if any) at least 12 hours before your session.
- You can use any models, tools, or frameworks you like — tell us why you chose them.

- Two appendices follow. **Appendix A** is a set of sample customer inquiries — treat them as the data you're designing for; if your demo runs on them, even better. **Appendix B** describes how Franklin sounds — design your drafts to match this voice.
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Appendix A — Sample customer inquiries (anonymized)

Use these as representative of what comes into our support inbox. They deliberately span easy wins, escalation cases, ambiguous requests, and compliance-sensitive topics. A good design handles them differently — some should produce a confident draft, some should produce a cautious draft, and some should produce *no draft and a routed escalation*.

Most examples are written in Danish, just like in reality — the bulk of our customers write to us in Danish, with the occasional English message. Your system needs to read and reply in Danish, not just English.

1 — Simple, well-documented

Hej Franklin

Hvordan ændrer jeg den email-adresse der er knyttet til vores konto? Jeg kan ikke finde indstillingen nogen steder under min profil.

På forhånd tak!

Med venlig hilsen Rasmus Holm Nordic Gear ApS

2 — Fee question (accuracy-sensitive)

Hej

Jeg kan se en postering på vores kontoudtog på 25 kr med teksten "service fee" i denne måned. Det havde vi ikke regnet med — kan I forklare hvad det dækker over, og er det noget vi kan få refunderet?

Tak for hjælpen.

Venlig hilsen Mette Bjerregaard Økonomiassistent, Brandt Logistik

3 — Account-specific, needs data + caution

Hej

Vores overførsel på 4.000 kr står stadig som "pending" to dage efter den blev sendt. Den skulle være gået igennem med det samme. Vi har brug for at pengene er fremme i dag, så det haster lidt — kan I se hvad der er galt?

Bedste hilsner Anders Toft CEO, Toft & Co

4 — KYC / compliance-sensitive

Hej Franklin

Nu har I bedt mig uploade mit pas for TREDJE gang, og nu er vores konto oven i købet blevet spærret. Vi har været kunder i to år. Det her er simpelthen ikke i orden — hvad foregår der, og hvornår kan vi bruge kontoen igen?

Mvh Camilla Sørensen Direktør, Sørensen Interiør

5 — Dispute / potential fraud

Hi

There are three transactions on our card from last night that we did not make — around 1,800 in total. Nobody on our team was using the card, we were all asleep. Please block the card immediately and reverse these charges.

Best regards, Daniel Friis Founder, Friis Studio

6 — Angry, churn-risk, vague

Hej

Det her er den dårligste app jeg nogensinde har brugt. Ingenting virker, og jeg spilder min tid. Hvis ikke der er nogen der rent faktisk hjælper mig nu, skifter vi til en konkurrent.

Thomas Lund Lund Byg

7 — Feature request (no real answer in docs)

Hej Franklin

Et hurtigt spørgsmål: understøtter I tilbagevendende automatiske overførsler til en ekstern bankkonto endnu? Og hvis ikke — er det noget der er på jeres roadmap?

På forhånd tak.

Venlig hilsen Sofie Krogh Finance Lead, Krogh Media

8 — Ambiguous / underspecified

Hej

Det virker ikke. Kan I fikse det?

Mvh Jonas

9 — Borderline financial advice (should not be answered directly)

Hej

Vi har omkring 50.000 kr stående på vores Franklin-konto lige nu. Synes I vi skal flytte dem over i jeres opsparingsprodukt, eller er det bedre at investere dem et andet sted? Hvad ville I selv gøre?

Bedste hilsner Henrik Dam Partner, Dam & Vinter

10 — Multi-part, mixed urgency

Hej Franklin

To ting: (1) vi er blevet trukket to gange for den samme betaling her til morgen, og (2) jeg vil også gerne høre hvordan jeg tilføjer en ekstra bruger til vores konto. Det vigtigste er at få styr på dobbelttrækket først.

Tak!

Med venlig hilsen Louise Bach Bogholder, Bach Retail ApS

These samples are illustrative and anonymized; amounts and details are fabricated. Replace or extend with your own anonymized tickets and a help-center excerpt before sending to candidates, so their demo runs on your real product surface.

Appendix B — How Franklin sounds (voice & tone)

Distilled from real Franklin support replies. A good drafting system should produce replies in this voice — and recognize the moments where the right move is a careful, human-reviewed message rather than a breezy one.

The short version

Franklin sounds like **a helpful, human colleague — not a bank**. We're warm, informal, and on a first-name basis with customers. We take genuinely complex things (KYC, funding, compliance, failed payments) and explain them simply and calmly, always making the next step obvious. We're on the customer's side, even when the process is bureaucratic.

If a draft sounds stiff, corporate, or like a legal disclaimer, it's wrong. If it sounds like a smart, friendly person who's clearly got your back, it's right.

Core characteristics

Warm and personal. We open with **Hej [first name]** and sign off as a real person — **De bedste hilsner, Caroline fra Franklin** — not "Kind regards, The Support Team." Customers are spoken to as people we know, not ticket numbers.

Informal, but competent. Conversational and relaxed (**Hejsa!, Dejligt at høre!, Sig endelig til hvis du har spørgsmål**), while still clearly in command of the details. Friendly does not mean vague.

Simple over technical. We translate financial and compliance complexity into plain language. When something has multiple steps (e.g. identity verification), we break it into a numbered checklist with the key actions in **bold**, and tell the customer exactly what to do — and what to avoid. Clarity is the whole point.

Reassuring and proactive. We close loops and take ownership: **Når ovenstående er klaret, sørger jeg for at du får adgang 🙌**. We tell people what we'll handle so they don't have to worry about it.

On the customer's side. Even with partners and back-office processes, we advocate for the customer and push back on needless friction — e.g. questioning why a customer should have to send a full bank statement when simpler details would do. We carry the burden, not the customer.

Light, natural emoji. A few warm emoji are part of the voice — 😊 🙌 🙌 — used to soften and add warmth, never decoration. Roughly one per message, in the greeting or the sign-off. Never in genuinely serious moments (fraud, frozen funds, money at risk).

Bilingual by default. Danish is our primary voice; we switch fluently to English the moment the customer writes in English. Mixed sign-offs like **Med venlig hilsen / Best regards** are normal. Always match the customer's language.

How the tone flexes by situation

- **Simple how-to questions** → friendly, quick, confident. Get them the answer and offer to help further.
- **KYC / onboarding / document requests** → extra clear and structured (numbered steps, bold actions), patient, and apologetic about friction we're imposing. These are where simplicity matters most.
- **Money-at-risk or upset customers** (failed payments, frozen accounts, possible fraud) → still warm, but calmer and tighter. Drop the emoji, acknowledge the worry, be concrete about what happens next and by when. Never breezy.
- **Anything sensitive or uncertain** (compliance judgment calls, financial advice, account-specific facts we can't confirm) → we don't wing it. We're honest about what we can and can't do and route it to a human.

Signature phrases (real Franklin)

- Greeting: **Hej [navn] · Hejsa! · Hej [navn]** 🙌
- Warmth: **Dejligt at høre! · Det lyder som en rigtig god idé** 😊
- Guiding: **Vær opmærksom på: (+ bullets) · Step 1 ... Step 2 ...**
- Ownership: **... så sørger jeg for at du får adgang** 🙌
- Open door: **Sig endelig til hvis du har spørgsmål** 😊
- Sign-off: **De bedste hilsner, Caroline fra Franklin · Med venlig hilsen / Best regards**

Quick do / don't

Do: use the customer's first name · explain the *why* behind a step · break complex asks into a checklist · sign as a named person · match the customer's language · keep it short.

Don't: sound like a bank or a legal notice · hide behind jargon or passive voice · over-apologize or grovel · use emoji when money or trust is on the line · make account-specific claims (balances, fees, refund status) the draft can't actually verify · give financial advice.